

Controlled wetting/dewetting through substrate vibration-assisted spray coating (SVASC)

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Abstract We have recently developed a novel spray-coating method, called the “substrate vibration-assisted spray coating” (SVASC) (Zabihi and Eslamian in *J Coat Technol Res* 12:711–719, ²⁰¹⁵), in which ultrasonic vibration is imposed on the substrate to improve the spray-on film and coating characteristics. In that work, the SVASC method was introduced, and its effectiveness on the uniformity and electrical conductivity of PEDOT:PSS films, used in emerging solar cells, was demonstrated. The present work reports unprecedented results on the effect of the ultrasonic vibration power on wetting/dewetting of PEDOT:PSS films. It is observed that, while the application of a low-power ultrasonic vibration (LPUV) improves the surface wetting and film coverage through improving droplet spreading and coalescence and repairing of the dewetted areas, a high-power ultrasonic vibration (HPUV) promotes dewetting, resulting in less coverage and the formation of a nonuniform film. The improved wetting due to the application of the LPUV has opened a window of opportunity for the fabrication of intact thin films and related thin film devices. On the other hand, the controlled dewetting process due to the application of the HPUV may have novel applications in template fabrication and self-assembly in nanotechnology. Here, we also study the effects of the application of multipass spraying compared with single-pass spraying strategy, and the application of using two co-solvents on PEDOT:PSS film characteristics. The results confirm that the utilization of isopropyl alcohol (IPA) as a co-solvent added to the PEDOT:PSS precursor solution improves the surface wettability and film coverage,

compared to the films made using demethylformamide (DMF), as the co-solvent.

Keywords Substrate vibration-assisted spray coating (SVASC), Ultrasonic substrate vibration, Wetting, Dewetting, PEDOT:PSS, Solution-processed solar cells, Perovskite solar cells

Introduction

Crystalline silicon-based and some inorganic thin-film solar cells, such as amorphous silicon solar cells, have been commercialized and used in specialty applications for decades. However, the high fabrication cost, high temperature, and time-consuming fabrication procedures hinder the widespread application of the solar energy through the photovoltaic effect. Emerging thin-film solar cells have been developing in the past years to address the cost issue associated with crystalline silicon-based and inorganic thin-film solar cells. Most layers of the emerging solar cells, such as polymer and perovskite solar cells, can be fabricated through easy solution-based processes, such as spray coating.¹ Various versions of spray coating have been used in large scale in painting, automotive, pharmaceutical, agricultural, cosmetic, textile, and food industries,² with the objective of providing a uniform protective or decorative deposition with the best deposition effectiveness (lowest wastage and energy consumption).³ In the laboratory scale, spray coating has been introduced to fabricate almost all layers of small-area solution-processed solar cells. Large-scale roll-to-roll manufacturing of future solar cells with low wastage of materials in a fast and cost-effective process is realizable by spray coating, whereas spin coating, the conventional lab scale fabrication method for solution-processed thin films, is applicable to small scales and batch processes only. Owing to the aforementioned

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tioned potentials, in recent years, spray coating has been widely employed for the fabrication of thin layers of solution-processed solar cells.⁴ In a typical spray coating process, a solar cell material is dissolved or dispersed in a proper solvent, then it is atomized to generate tiny spray droplets carried onto a substrate by a carrier gas. Spray droplets then impact the substrate, whereby they spread, coalesce, overlap with previous droplets, and finally dry and leave a thin solid film behind.⁵ Spray coating can be performed in a single spray pass, where the film forms as the spray nozzle moves above the substrate, or in multiple spray passes, to improve the surface coverage and perhaps the film uniformity in some cases.

Despite the potentials, the mechanism of spray coating for the fabrication of thin films used in solar cells is not well understood, due to the complex and stochastic nature of sprays, which makes the fabrication of thin and smooth films quite challenging.⁵ To overcome the challenges, efforts have been made to improve various characteristics of spray-on organic and perovskite solar cells.^{4,6–8} A thin film with poor coverage and numerous defects sandwiched between other layers may cause undesired contact of its adjacent layers to one another (reduced shunt resistance), thus discounting the device efficiency. The capacity to fabricate defect-free, continuous, and smooth thin solid films is central to the successful performance of photovoltaic and other optoelectronic devices incorporating such films. Undesirable tears in the film before complete drying may occur due to liquid surface instabilities, substrate contamination, etc.⁹ Some works have comprehensively reviewed the dewetting phenomena of liquid thin films.^{9–12} In the past two decades, efforts have been made to understand the dewetting mechanism, summarized as follows^{9,13}: (1) dewetting due to heterogeneous hole nucleation in which impurities and defects in liquid film and substrate initiate hole generation; (2) dewetting by thermal nucleation or homogeneous nucleation caused by temperature gradients; (3) spinodal dewetting, initiated by long-range molecular forces in the liquid and interfaces, and (4) autophobic dewetting, which is driven by entropy. We have recently identified another sort of film dewetting caused by precipitation and crystallization in liquid solutions. Although dewetting may have some novel applications, such as creation of nanoislands of iron on the surface to be used as a template for the synthesis and growth of carbon nanostructures,¹⁴ in most cases, such as coating and film formation applications, it is detrimental to the film characteristics, and has to be avoided. Several approaches may be adopted to suppress dewetting, such as careful removal of impurities and contaminants from the solution and the surface, substrate surface modification, and inclusion of surfactants into the solution, to alter the chemical and intermolecular forces responsible for dewetting. Droplet spreading and surface wetting is also controlled by dynamic forces acting on the film.¹⁵ A tangible example is the

positive effect of centrifugal forces in spin coating that helps a uniform film to form. In spray coating, the dynamic forces, such as the momentum associated with the impinging droplets, enhance spreading and wetting, but once these forces are removed, the film may tend to retract and reach the equilibrium condition, and thus becomes susceptible to dewetting, if the film is still wet. Imposing ultrasonic vibration on the substrate, the idea and focus of this work, is a dynamic force that can promote wetting, if its vibration power or amplitude is well controlled. The vibration does not have to be ultrasonic, however, high frequency and low wavelength vibrations can more effectively cause droplet spreading and coalescence, and dampen surface waves. The concept of imposing vibration, shaking, or moving of a substrate has been studied on droplets, but not on films and coatings. It has been observed that substrate or droplet vibration results in some interesting phenomena, such as droplets moving uphill,¹⁶ and improved droplet spreading.^{15,17} The phenomenon of impact of a liquid jet on a fast-moving substrate has been studied, as well.^{3,18,19} An extensive review of the reports on the effects of substrate vibration and movement on sessile and impinging droplets and jets is given in our previous work.²⁰

Inspired from the positive effects of substrate vibration on droplet spreading, we have recently introduced and employed a novel spray coating method, i.e., the substrate vibration-assisted spray coating (SVASC) to fabricate thin films and coatings, and have demonstrated its outstanding effects on improving the film wetting, surface coverage, and film functionality.^{20,21} Zabihi and Eslamian²¹ applied ultrasonic vibration on a paper substrate during spray coating of a food dye and observed a more uniform thin film, when the substrate is vibrated using a low-power ultrasonic transducer. They also implemented this new technique on poly(3,4-ethylenedioxythiophene):poly(styrene sulfonic acid) (PEDOT:PSS) thin films and showed that substrate vibration brings about a significant enhancement in the electrical conductivity of PEDOT:PSS films.²⁰ PEDOT:PSS is readily dispersed in water and other solvents; has hole transporting ability, good transparency, and high work function; and thus is used in optoelectronic devices and emerging solar cells. PEDOT:PSS is composed of PEDOT, a conducting polymer, and PSS, a counter-ion and charge compensator part. PEDOT is insoluble in most solvents, but it can be dispersed in water using PSS.²² In an aqueous dispersion, PEDOT grains, the hydrophobic and conducting part, are encapsulated in PSS, which acts as a hydrophilic and insulating shell.²³ In polymer or perovskite solar cell configurations, PEDOT:PSS has been fabricated widely by solution-processed techniques, such as spin coating^{22,23} and spray coating.^{4,24,25}

Although our previous works on the application of SVASC using a low ultrasonic vibration amplitude or power^{20,21} revealed the effectiveness of imposing ultrasonic vibration on the substrate for the formation

of smooth and uniform films, the effects of the vibration power were not studied and some parameters, such as the film coverage and roughness, were investigated qualitatively. It is expected that imposing vibrations with a power or amplitude beyond a threshold would promote film dewetting. The main goal of this article is therefore to examine the aforementioned hypothesis on the effect of the vibration power on wetting/dewetting. In particular, the effects of varying the vibration power on PEDOT:PSS film characteristics are studied, using two co-solvents, 2-propanol (IPA), and dimethylformamide (DMF). IPA and DMF have different surface tensions, and their addition to the PEDOT:PSS aqueous solution modifies the contact angles of PEDOT:PSS droplets placed on the substrate. A solution with smaller surface tension (IPA-added solution) is favorable for the fabrication of a uniform thin film with higher coverage, due to better wettability.^{4,26} In this work, the effects of utilizing single and multiple spray passes on wetting characteristics are studied as well.

Experimental

Materials and methods

Glass substrates with dimensions of 20 mm × 20 mm were cleaned in an ultrasonic bath for 15 min, using detergent, acetone, and deionized water, and dried in a vacuum furnace for 10 min, and in an UV-Ozone cleaner for 15 min. PEDOT:PSS pristine aqueous solution used here (Sigma-Aldrich, USA) consisted of 0.5 wt% PEDOT and 0.8 wt% PSS, dispersed in water. It was further diluted by dissolving it in either IPA or DMF organic solvents. The diluted solutions were prepared in a 2:1 volume ratio (40 mL of pristine PEDOT:PSS solution mixed with 20 mL of IPA or DMF) and were stirred with magnetic stirrer before spraying. The solutions were sprayed with spray pyrolysis equipment (Holmarc Opto-Mechatronics Pvt. Ltd., Model HO-TH-04, India). Ultrasonic vibration is used for two purposes, i.e., first to atomize the solution using an ultrasonic nozzle, and then to vibrate the substrate to affect the film wetting. The solutions were pumped from a 50 mL container to an ultrasonic spray nozzle. The ultrasonic nozzle was used for atomization of the solution into droplets with the average diameter of 50 μm. More details about the setup may be found elsewhere.²⁰ A second ultrasonic transducer was installed in a metal box and used as a platform to hold and transfer ultrasonic vibrations to glass substrates to improve the coating characteristics. The substrate ultrasonic vibration signals were generated using a signal generator (Clangsonic, SONOCLG MS600, China) with a controllable power from 5 to 50 W, and a fixed frequency of 40 kHz. Figure 1 schematically illustrates the SVASC experimental setup.

In spray coating, several process parameters, such as solution concentration, substrate temperature, number

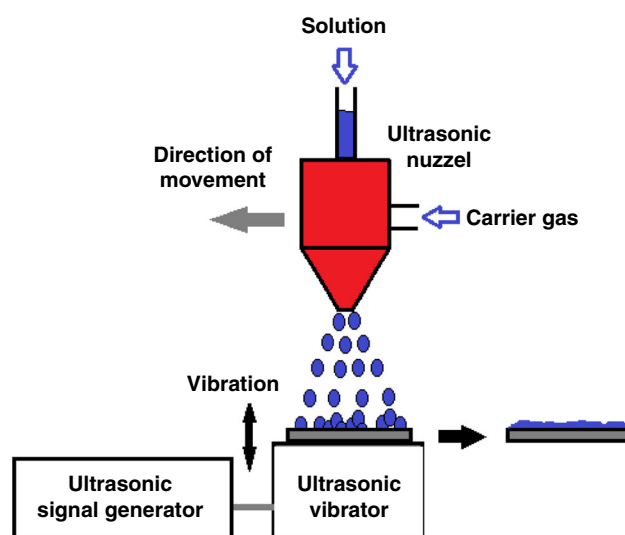


Fig. 1: Schematic of the experimental setup: substrate vibration-assisted spray coating system (SVASC)

of spray passes, distance of the ultrasonic nozzle tip to substrate, liquid flow rate, nozzle air pressure, and nozzle speed with respect to the substrate, influence the film characteristics. In the SVASC, in which an ultrasonic vibration is imposed on the substrate, additional parameters are involved, including the power, amplitude, and frequency of the substrate vibration. Here, except the number of spray passes, solvent type, and the vibration power or amplitude, all other parameters are kept constant, given that we are mostly concerned about the effects of vibration power on wetting/dewetting. All experiments were conducted in ambient temperature (~25°C); the distance of the substrate to nozzle tip was kept constant at 60 mm, the liquid flow rate was set to 5 mL/min, the nozzle air pressure was set to 3 bar, the solution concentration was 0.0086 g/mL, and the nozzle tip velocity was preset to 25 mm/s. In the spray coater, the nozzle tip horizontally moves over the substrate placed on the ultrasonic transducer box.

Experimental runs were carried out using two solvents in various substrate modes, including the substrate stationary mode (zero vibration power), low-power ultrasonic vibration (LPUV) at 5 W (vertical substrate displacement of 8.7 μm at 40 kHz), and high-power ultrasonic vibration (HPUV) at 50 W (vertical substrate displacement of 57.7 μm at 40 kHz). Three spraying strategies were adopted including the single pass, two passes, and four passes. Table 1 lists different experimental runs, based on the type of the solution, power of the ultrasonic vibration imposed on the substrate, and the number of the spray passes. Besides spray coating experiments, two PEDOT:PSS samples were also fabricated by spin coating for each solvent for comparison with the spray-on samples. The samples were spin coated in a glove box using spinning speed of 2000 rpm for 1 min. Each spray or spin coated sample

Table 1: Spray coating experiments performed to fabricate PEDOT:PSS films on glass substrates using two solvents

Co-solvent = DMF			Co-solvent = IPA		
Run	# of passes	Vibration power (W)	Run	# of passes	Vibration power (W)
1	1	0	10	1	0
2	1	5	11	1	5
3	1	50	12	1	50
4	2	0	13	2	0
5	2	5	14	2	5
6	2	50	15	2	50
7	4	0	16	4	0
8	4	5	17	4	5
9	4	50	18	4	50

was allowed to dry at the ambient temperature for 1 min, and then stored in a sealed glass container for characterization.

In addition to the fabrication of the aforementioned PEDOT:PSS films, in order to examine the effect of the substrate ultrasonic vibration on wetting/dewetting at the interfaces of layers with particular application in thin-film solar cells, light harvesting perovskite thin films were spray coated on PEDOT:PSS thin films, deposited on fluorine-doped tin oxide (FTO) glass (Sigma-Aldrich, USA). For this purpose, the precursor solutions of perovskite were mixed at 70°C and filtered by a 30- μ m PTFT filter (Cole-Parmer, USA). Resultant filtrate was then loaded into a gas-assisted syringe pump mounted on the spray pyrolysis machine described above and deposited on PEDOT:PSS films in both vibrating at 5 W and stationary modes. Temperatures of the substrate and syringe pump were kept at 70°C. Details of apparatus and operating procedure were presented in our previous work.²¹ Perovskite films were deposited atop PEDOT:PSS films at solution flow rate of 400 μ L/min, air pressure of 3 bars, speed of nozzle of 5 mm/s, and nozzle to substrate distance of 6 cm. Perovskite reagents consisted of methyl ammonium iodide in 2-propanol (0.01 g/mL) and lead chloride in DMF (0.24 g/mL). The volume ratio of perovskite precursor solutions was set to 3:1 molar ratio of methyl ammonium iodide to lead chloride. All the chemical reagents and organic solvents were used in analytical grade and provided by Sima-Aldrich, USA. In total, 22 samples (2 spun-on and 18 spray-on PEDOT:PSS films and 2 spray-on perovskite films) were fabricated.

Sample characterization

To characterize the PEDOT:PSS films, several methods were employed. Confocal laser scanning micro-

scopy (CLSM, model LMS700, Zeiss, Germany), with both the optical mode (for 2D and 3D surface visualization), and the high-resolution laser mode (for roughness and thickness measurements, as well as visualization) was used. The optical images in a large field of view of 700 μ m $\times$ 525 μ m were taken to have a wide view of the surface patterns of spray-on films, and to calculate the percentage of coverage. The average droplet size at the nozzle tip is 50 μ m, and therefore, the employed field of view ensures that droplet and splat (dried impinged droplets) overlapping, possible coffee rings, and similar effects are visible. The covered areas of each sample (areas where an intact and noticeable film actually forms) were calculated based on a color contrast between the coated and uncoated areas. This was performed on six spots of the film randomly chosen on each sample, and then averaged to obtain the final surface coverage values. In the laser mode, the images were taken in three different magnifications (in the field of view of 650 μ m $\times$ 650 μ m, 130 μ m $\times$ 130 μ m, and 65 μ m $\times$ 65 μ m) to calculate the film roughness, given that the roughness of spray-on films is highly dependent of the size of the measurement area. Roughness values were also reported based on the average roughness values obtained from six random spots on each sample. Roughness was calculated using the “arithmetic mean derivation of all surface height values” method which was done by the software of the CLSM. At higher magnifications (lower field of view), the images were taken from locations on the substrate where a film was actually formed and observed by inspection. In other words, the roughness values associated with a small field of view represents the actual film roughness, whereas, those obtained from larger fields of view take into account the dewetted areas as well.

Scanning electron microscopy (SEM, Model S-3400N, Hitachi, Japan) was also employed for high resolution surface characterization of selected samples (PEDOT:PSS in DMF solvent with four spray passes and the perovskite thin films sprayed on PEDOT:PSS films). Nanoscale resolution of surface profile of selected samples (PEDOT:PSS in IPA solvent with four spray passes) was characterized by an atomic force microscope (AFM, Veeco Nanoscope Instrument, USA) in the tapping mode.

The solution droplet contact angles and the interfacial tension of liquid solutions with respect to the glass substrate were measured at room temperature. The contact angle was measured by an optical contact angle measurement instrument (Dataphysics, model OCA 15, China). A surface tension meter (KRÜSS, model K100, Germany) was used to measure the interfacial tension. The amplitude of the vertical displacement of the substrate caused by the imposed ultrasonic vibration was measured by a vibrometer (Cheersonic Ultrasonic Co., China). The vertical displacement of the ultrasonic transducer at 5 W was measured to be 8.7 μ m and at 50 W it was measured to be 57.7 μ m. The viscosity of the pristine PEDOT:PSS aqueous

solution at room temperature was measured and reported in our previous work²⁷ and is 0.9 mPa.s close to that of water. Also, the viscosity of the IPA- and DMF-treated PEDOT:PSS solutions remains close to that of water. Therefore, the flow of the precursor solution impinging on the substrate and making a liquid film is Newtonian.

Results and discussion

In spray coating, spray droplets impinge on a substrate followed by droplet spreading and coalescence, solvent evaporation, solute precipitation, etc. Process parameters such as the solution properties, substrate wetting behavior, and temperature may affect droplet dynamics and the coating process.¹ Here, the substrate temperature is kept constant, while the effects of varying the solution physical properties on wetting behavior are investigated employing IPA and DMF co-solvents mixed with pristine PEDOT:PSS solution. In order to compare the physical properties of these two solutions, contact angles and interfacial tensions of the solutions with respect to the glass substrate were measured. Table 2 shows the measured contact angles of PEDOT:PSS solution droplets mixed with DMF and IPA, as well as the interfacial tensions between these solutions and the glass substrate. Low contact angle and interfacial tension of PEDOT:PSS mixed with IPA droplets placed on glass substrates, compared to those of PEDOT:PSS mixed with DMF predicts better wettability of the solution mixed with IPA compared to that of DMF. DMF co-solvent was reported to increase the conductivity of spray-on PEDOT:PSS films²⁴ (a property related to the film microstructure), although this solution has a high interfacial tension (low wetting). After droplet impingement on the substrate, solid-liquid interfacial tension has an important effect on splashing, spreading, and rebounding of droplets.² It has been reported that liquids with high interfacial tensions, such as PEDOT:PSS aqueous solution mixed with DMF, cannot wet the substrate properly.²⁸ In this case, wetting is poor, and due to slow solvent evaporation in ambient temperature, impinged droplets may merge forming larger droplets leading to a rougher surface. In contrast, PEDOT:PSS aqueous solution mixed with IPA exhibits improved wettability according to previous studies^{28,29} and data of Table 2. Therefore, a more uniform spray-on thin film is

expected to form when aqueous solution of PEDOT:PSS diluted with IPA is used.⁶ This effect will be further investigated in the following sections.

In spray coating, substrate vibration usually helps spreading of impinged droplets; when the rims of two adjacent droplets approach one another, the process of droplet coalescence occurs by creating a bridge between the two droplets, where the bridge grows and the shape of the combined droplet is rearranged from elliptical to circular shape.³⁰ Application of a higher spray flow rate, multiple spray passes, and substrate vibration increases the rate of droplet coalescence and the likelihood of obtaining a flat liquid film, and subsequently an intact solid film. Application of sprays with multiple deposition passes has been suggested in most of the recent works due to the improvements observed in film uniformity and coverage when multiple passes are employed.^{4,20,21,26,27} It has been also suggested that if ultrasonic vibration is imposed on the substrate, a single spraying pass may sufficiently produce a uniform film of PEDOT:PSS, if the flow rate is high enough.²⁰ Some other workers believe that although multiple spray pass strategy results in a better coverage, smoother films are obtained by a single-pass spraying strategy.^{1,27,31}

Figure 2 shows the large field of view (650 $\mu\text{m} \times 650 \mu\text{m}$) optical images of single-pass spray-on PEDOT:PSS films made using IPA (left column) and DMF (right column) solvents, at various substrate vibration conditions. The dark (blue) areas clearly show the formation of a rather thick film, although some areas of the substrate are covered by a light color, suggesting the presence of no film or a very thin film in such areas. Theoretically, it is possible to estimate whether the delivered solution to the substrate is enough to cover it up by a thin liquid film or not. An impinging droplet spreads, forming a disk with a final diameter (d) of 1.5–5 times the diameter of the impinging droplet diameter (D) upon impact. The disk diameter is a function of parameters, such as the impinging droplet Weber number, Reynolds number, surface wettability and roughness, and contact angle.¹ A mass balance between the incoming flow of the solution to the substrate and the deposited film results in the following equation for the minimum solution volume flow rate $q_{\min}$ required to form a liquid film by the precursor solution^{4,26}:

$$q_{\min} = \frac{2}{3} w_a u_N \left(\frac{D^3}{d^2} \right), \quad (1)$$

where w_a is the diameter of the wetted area on the substrate when the nozzle tip is in the stationary mode, and u_N is the speed of the nozzle tip. Equation (1) estimates the minimum required volume flow rate to cover up the substrate with a uniform continuous liquid film. In our experiments, the single spray pass flow rate is 5 mL/min, which is larger than the theoretical value obtained from equation (1), i.e., 2.2 mL/min assuming

Table 2: Physical properties of PEDOT:PSS aqueous solutions diluted with DMF and IPA

Solution	PEDOT:PSS in DMF	PEDOT:PSS in IPA
Interfacial tension (mN/m)	19.67	11.93
Contact angle (°)	12.0 $\pm$ 1.96	1 $\pm$ 0.50

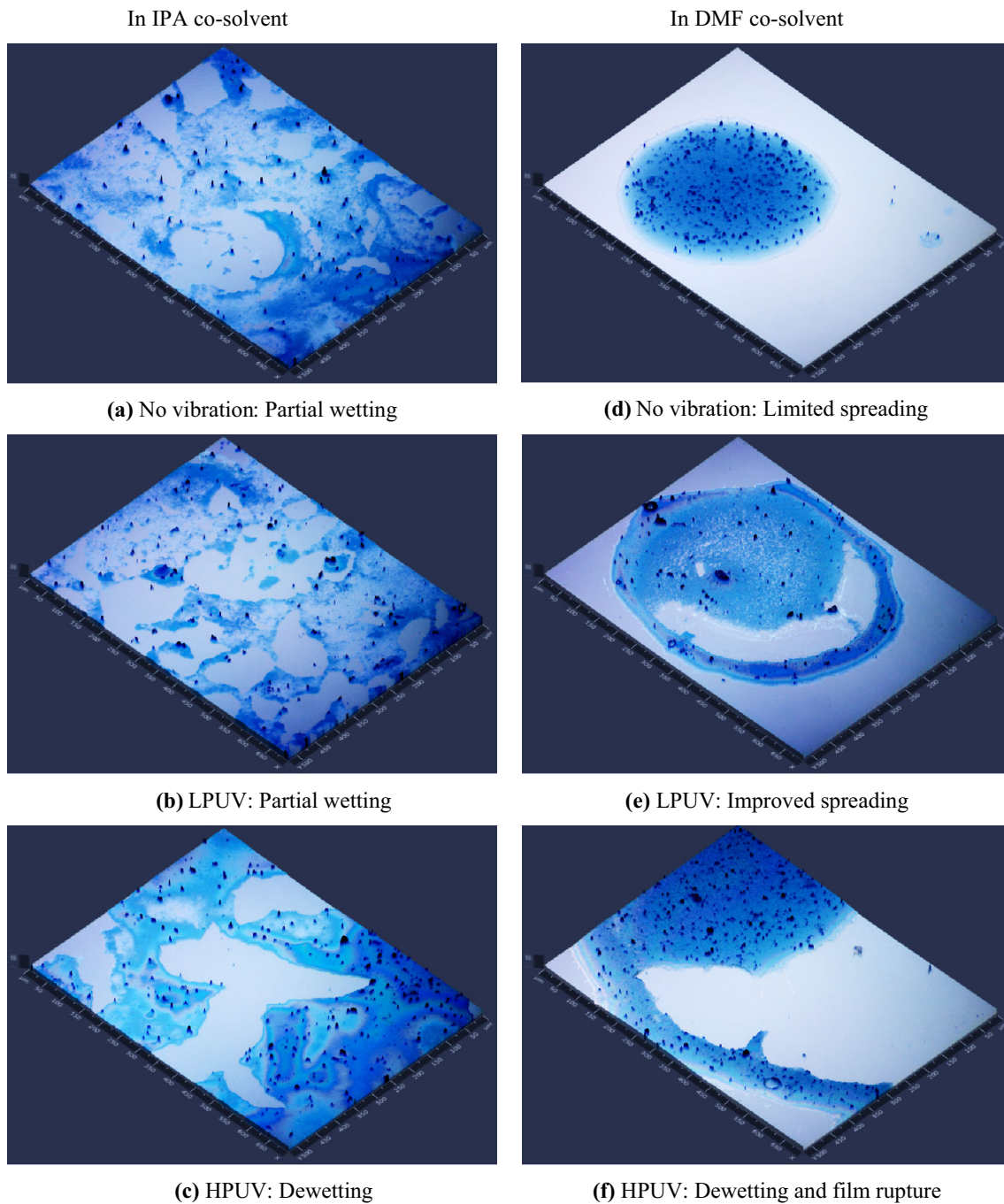


Fig. 2: Optical images of spray-on PEDOT:PSS films using IPA (left column) and DMF (right column), as the co-solvent in stationary (a, d), LPUV (b, e), and HPUV (c, f) cases. Single spray pass was applied

that $w_a = 0.1$ m, $u_N = 0.025$ m/s, $d = 75$ μ m, and $D = 50$ μ m (assuming a conservative spreading). Therefore, sufficient amount of solution has been delivered to cover up the substrate, although the solid film may be very thin in some areas due to the nonuniformities observed in spray-on films, or the liquid film may have ruptured due to the dewetting effects. Also, it is noted that the droplets are not uniformly distributed in the spray, leading to uneven delivery of solution to the substrate. The spray is

usually denser in the center or core than the edges. Therefore, even though a nanoscale film may have formed in light color areas of the substrate, it is safe to assume that the film is ruptured in such areas, due to dewetting or lack of sufficient amount of delivered solution.

Following the foregoing discussion, Fig. 2a (Run 1; single spray pass, no vibration, IPA as the co-solvent) shows that an uneven film of PEDOT:PSS has formed, where several uncovered or dewetted areas are

observed. In this case, it is obvious that the high wettability of the IPA solution has resulted in spreading of droplets, but a complete, uniform, and effective wetting and coverage is absent, and the film is nonuniform and disjointed. For the same co-solvent, using an LPUV imposed on the substrate, as shown in Fig. 2b, leads to slightly increased coverage, whereas a high power vibration has a clear adverse effect on the film continuity, causing a systematic rupture in the film (Fig. 2c). Changing the co-solvent from IPA to DMF (right column) causes different film morphologies, due to differences in physical properties of the two solutions, such as different interfacial tensions and contact angles, as reported in Table 2. After spraying of the solution containing DMF on the stationary substrate with no vibration, while, overall, sufficient amount of liquid has been provided to cover up the entire surface, the impinging droplets do not favorably wet the surface and only cover parts of the glass substrate (Fig. 2d). It clearly shows that DMF-based solution is less wetting compared to IPA-based solution, which is in good agreement with their measured physical properties. Application of a LPUV enhances spreading, but the film is still ruptured (Fig. 2e). Similar to the previous case, the application of the HPUV results in a decrease in the coverage (Fig. 2f). Therefore, application of the LPUV improves spreading and wetting, and application of the HPUV promotes dewetting.

The effect of the substrate vibration and its power on the film morphology and wetting in the experiments performed using one spray pass was found inconclusive, because insufficient amount of solution was delivered to the substrate to make a uniform and continuous liquid film, and subsequently a continuous solid film. The experiments performed with two and four spray passes can therefore better substantiate the effect of the substrate vibration on wetting/dewetting. Figure 3 shows the optical images of PEDOT:PSS films made from a solution with IPA (left column) and DMF (right column) as the co-solvent, using a double-pass spraying strategy. The spray droplets of the second pass impinging atop the first pass are meant to improve the surface wetting, coverage, and film uniformity. Visual inspection of the films of Fig. 3, made under various conditions, shows that the application of the LPUV improves the wetting, whereas application of the HPUV deteriorates the surface wetting; this is because the high power vibration makes the film unstable, through vigorous perturbation of the film. Changing the co-solvent from IPA (Fig. 3, left column) to DMF (Fig. 3, right column) leads to similar observations, although for the DMF case, the surface wetting and coverage is discounted, due to the high interfacial energy and contact angle of the DMF-based PEDOT:PSS solution (c.f. Table 2), which leads to limited droplet spreading on the substrate. The effects of the substrate vibration power on surface wetting are best documented when four-pass spraying strategy is adopted, in which case the substrate is reasonably covered by the liquid precursor (Fig. 4). In the case of

using IPA co-solvent (left column), the solution wets the entire surface, without using imposed ultrasonic vibration (Fig. 4a), and also with using the LPUV (Fig. 4b), but the LPUV produces a more uniform film, because of the moderate shaking effect of the LPUV, which smoothens the film by filling the valleys. Using the HPUV in this case, however, excessively disturbs the film, and breaks it up into separate islands (Fig. 4c). Application of four deposition passes for the solution of PEDOT:PSS in DMF (right column) shows uncovered areas indicating that obtaining a complete coverage with the solution containing DMF entails the application of additional spray passes or delivered liquid. The right column of Fig. 4 shows the effect of increasing the vibration power on the coating of the surface from zero to LPUV, and HPUV, and the way that the HPUV causes a clear-cut dewetting.

Figures 2, 3, and 4 provide a qualitative investigation of the effect of co-solvent, the number of deposition layers, and the power or amplitude of the imposed vibration on the film wetting/dewetting. Figures 5 and 6 provide quantitative analyses on the effects of the above-mentioned parameters on the film coverage and roughness. The surface coverage results are shown in Fig. 5, in agreement with the aforementioned qualitative discussion provided for Figs. 2, 3, and 4, i.e., (1) surface coverage of PEDOT:PSS film increases with increasing the number of spray passes; (2) the coverage of a PEDOT:PSS film made using IPA co-solvent is significantly higher than that of DMF-based film; (3) the coverage of a PEDOT:PSS film made using IPA co-solvent typically increases when an LPUV is imposed on the substrate, and decreases when an HPUV is employed. In the case of using IPA co-solvent, a full coverage was achieved when four deposition layers were employed, no substrate vibration, and also LPUV modes, whereas the HPUV dewetted the surface. In this case, the best coverage at two spray passes is achieved when the LPUV is applied, while in the single-pass spraying case, vibration has no effect, because the delivered liquid is not sufficient to cover up the substrate, effectively. Figure 5 also shows the coverage of spun-on films made using identical precursor solutions to those used for spraying experiments for comparison. Spun-on PEDOT:PSS film made using IPA shows a perfect coverage. For spray-on samples made using DMF-based precursor solutions (Fig. 5b) only partial coverage was achieved, even using four deposition passes. Spun-on films, made using DMF, had a higher coverage compared to spray-on films, although full coverage was not achieved. This was explained before using the data of Table 2. The effects of the imposed vibration on the coverage depend on the number of spray passes, and the co-solvent (Fig. 4). With IPA as the co-solvent, application of a LPUV is beneficial to better coverage, while the application of a HPUV is detrimental. When DMF is used, the response of the film to vibration is quite mixed; in the four-pass spraying strategy, increasing the vibration power from zero to high at 50 W

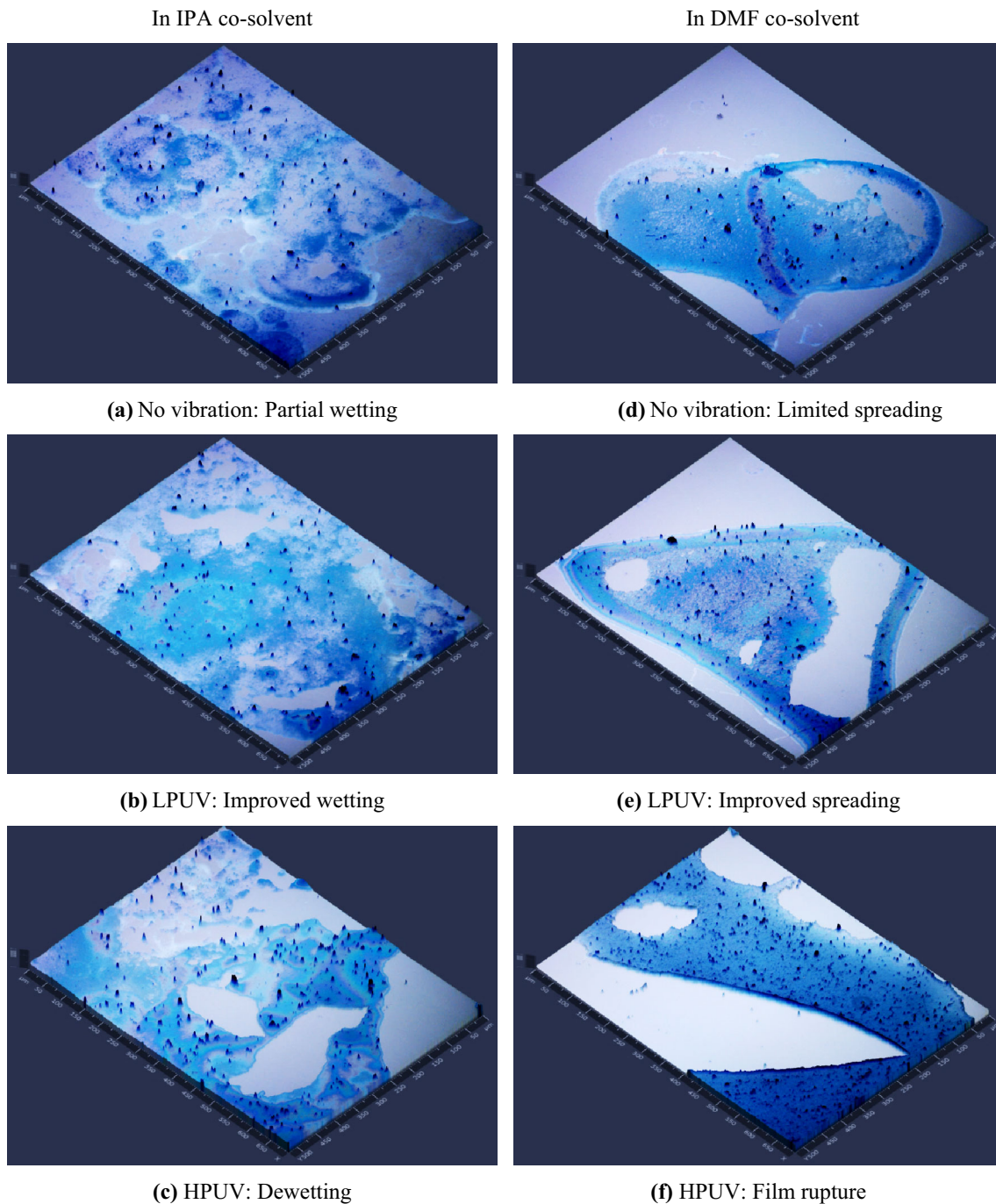


Fig. 3: Optical images of spray-on PEDOT:PSS films made using IPA (left column) and DMF (right column) in stationary (a, d), LPUV (b, e), and HPUV (c, f) modes. Two sequential spray passes were applied

causes increased dewetting and decreased coverage, while for one and two spray passes, the effect of the substrate vibration on the surface coverage is not conclusive.

In addition to surface coverage, roughness is another parameter that varies significantly with the ultrasonic substrate vibration; it also changes with changing the number of spray passes. The film roughness may

influence the functionality of the films used in a thin film device, such as a solar cell. Roughness values of the samples made in this study were calculated by the laser mode of the CLSM, for three fields of view. Unlike the spun-on films that are quite uniform, spray-on films are usually nonuniform, and therefore the roughness of spray-on films may depend on the size and choice of the measurement area. If the measure-

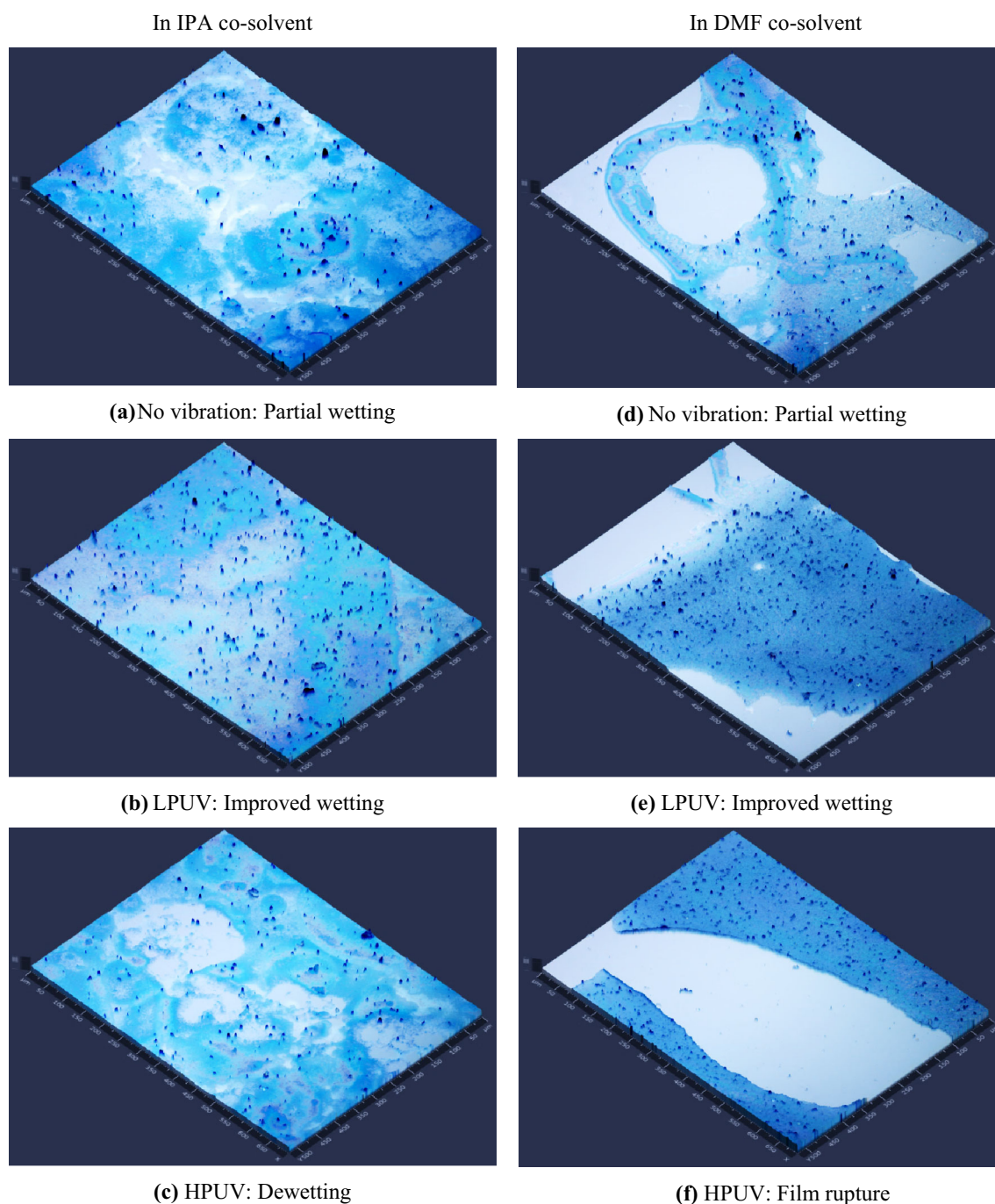


Fig. 4: Optical images of spray-on PEDOT:PSS films in IPA (left column) and in DMF (right column) in stationary (a, d), LPUV (b, e), and HPUV (c, f) modes. Four sequential spray passes were applied

ment area contains droplet boundaries, and protruded edges of the coffee rings, the measured roughness will be larger, whereas the roughness associated with a small local area within an impinged droplet is expected to be smaller. Following this argument, the roughness values of spray-on PEDOT:PSS films fabricated using IPA (left column), and DMF (right column) in three different scales, i.e., three different measurement

areas, are illustrated in Fig. 6. Moreover, the graphs contain roughness values of spun-on PEDOT:PSS films made using IPA and DMF, where it is found that the spun-on films are smoother than spray-on films. It is interesting to note that for all measurement areas, the roughness decreases when the LPUV is imposed on the substrate, and increases again when the vibration power is increased to the HPUV. Imposing the HPUV

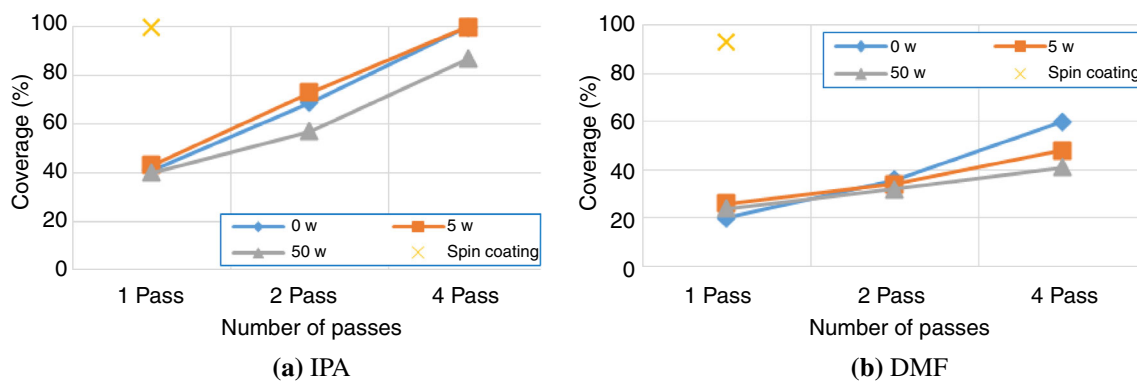


Fig. 5: Coverage of spray-on PEDOT:PSS films vs the number of spray passes, made at different conditions. (a) IPA-based precursor solution; (b) DMF-based precursor solution. The coverage of spun-on films is also shown, where the number of spray passes is irrelevant in this case

on the substrate in all cases yields higher roughness than those imposed by the LPUV, and in some cases, the roughness caused by the HPUV is even higher than the case without imposing vibration.

The effect of the number of the applied spray passes on the film roughness depends on the type of the co-solvent used, as well as the size of the measurement area. For instance, Fig. 6a shows how the roughness values change in a large view measurement area ($650 \mu\text{m} \times 650 \mu\text{m}$) with IPA as the co-solvent. In this graph, the single deposition pass has the highest roughness, for all substrate vibration scenarios, due to the existence of tears and dewetted areas in the film (c.f. Fig. 2). For two and four spray passes, the dewetted areas decline and in some cases vanish (c.f. Figs. 3, 4) resulting in a decrease in roughness. This behavior is different when the less wetting solution, treated with DMF, is used, and the results are inconclusive. One systematic observation is made for the smallest measurement area, i.e., ($65 \mu\text{m} \times 65 \mu\text{m}$), in which for both co-solvents and all three substrate vibration scenarios, an increase in the number of spray passes results in an increase in the local surface roughness of PEDOT:PSS films, due to the overlapping of the impinging droplets. For larger measurement areas, however, other effects, such as droplet boundary overlapping, and dewetting, affect the results. As another general observation, PEDOT:PSS films made using the precursor solution treated with the less wetting DMF co-solvent have higher roughness values for all the cases, compared to those films made using IPA, owing to the favorable physical properties of IPA compared to DMF (c.f. Table 2). Figure 6c shows that spray-on PEDOT:PSS films made under the LPUV condition, using IPA as the co-solvent, and using single or double spray passes have the lowest local roughness.

The effect of changing the power of the ultrasonic vibration imposed on the substrate is schematically depicted in Fig. 7. This figure shows the approximate film formation patterns, and wetting/dewetting behaviors for three cases, i.e., stationary or no-vibration

substrate (case 1), applying the LPUV on the substrate (case 2), and applying the HPUV on the substrate (case 3). The right column refers to the observations made based on the PEDOT:PSS films made using DMF as the co-solvent, while the left column corresponds to PEDOT:PSS films made using the more wetting co-solvent, i.e., IPA. The surface patterns in Fig. 7 do not necessarily indicate the surface waves, given that the ultrasonic waves in the conditions of these experiments are in millimeter and centimeter wavelength range and may be larger than the surface features shown. Therefore, the surface features are mainly due to the film roughness. It is also noted that the vertical displacement of the substrate due to the ultrasonic vibration is in the micrometer range, as reported earlier. The vertical displacement of the substrate may be regarded as the amplitude of the ultrasonic waves. These figures illustrate that low vibration indeed helps spreading, coalescence, and repairing of the dewetted areas, leading to the formation of a smoother and more uniform film, while the severe mechanical force due to the HPUV excessively perturbs the film resulting in film rupture and increased dewetting and the formation of a film with high roughness and low coverage.

To have a better view and understanding of the crystal or grain formation on the film surface, and to study the effect of the imposed ultrasonic vibration on grain formation, SEM images were taken for selected PEDOT:PSS samples, made using four spray passes in a precursor solution treated with DMF. Figures 8a–8c show the SEM micrographs of these films for three different vibration scenarios, i.e., stationary substrate, applying the LPUV, and the HPUV, respectively. Comparison of these images reveals the influence of the imposed ultrasonic vibration on the surface crystals or grain shapes, sizes, and population densities. The LPUV causes the creation of thinner and narrower crystals compared to the stationary state case, in which few small, round-shape crystals form. Increasing the vibration power to high values (HPUV) leads to the

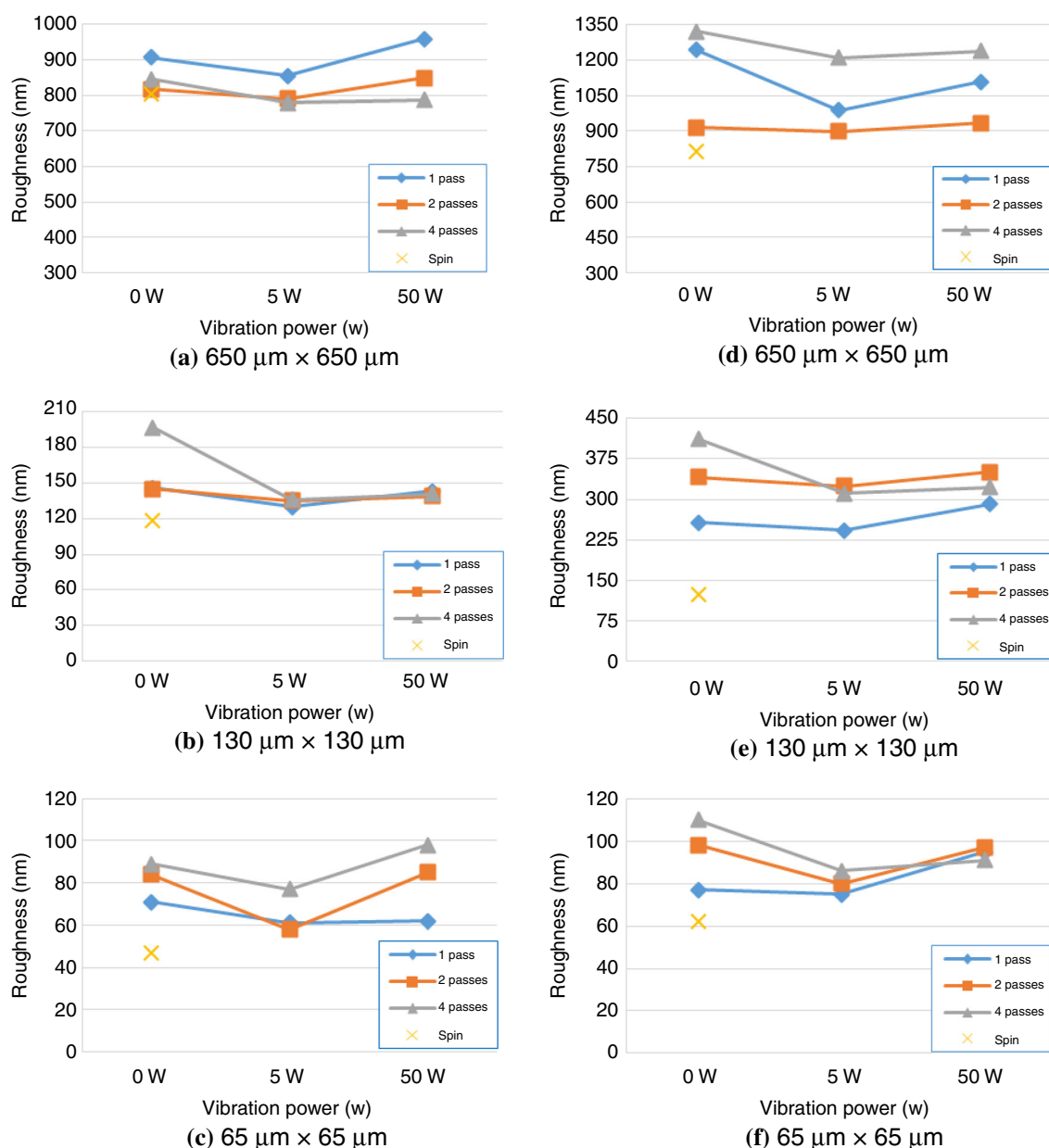


Fig. 6: Roughness values of spray-on and spun-on PEDOT:PSS films formed using IPA (left column) and in DMF (right column) in three different scales: (a, d) 650 $\mu\text{m} \times 650 \mu\text{m}$, (b, e) 130 $\mu\text{m} \times 130 \mu\text{m}$, and (c, f) 65 $\mu\text{m} \times 65 \mu\text{m}$

formation of thicker and wider crystals again, similar to those without using vibration, but with a higher population density. In other words, application of the LPUV smoothens the surface grains.

Three representative AFM height images of PEDOT:PSS thin films are shown in Fig. 9. These films were made using a precursor solution containing IPA as the co-solvent, and using four spray passes. AFM photos were taken to have a nanoscale sense of the effect of the substrate vibration on the morphology and surface topography of thin films. In the case of no vibration, some small grains are observed (Fig. 9a), typically observed in PEDOT:PSS films.²⁷ Application of the

LPUV results in surface refinement and elimination of the surface grains and protrusions (Fig. 9b), whereas the application of the HPUV results in the excessive agitation of the liquid film, and thus re-coarsening of the solid film (Fig. 9c).

One of the applications of the SVASC is to form uniform thin films used in thin film devices such as thin-film solar cells. Therefore, we investigated the effect of substrate vibration on wetting behavior of spray-on perovskite light harvesting thin films deposited on spun-on FTO/PEDOT:PSS films in both LPUV and stationary modes. Figure 10 clearly shows a remarkable improvement in coverage, roughness, and mor-

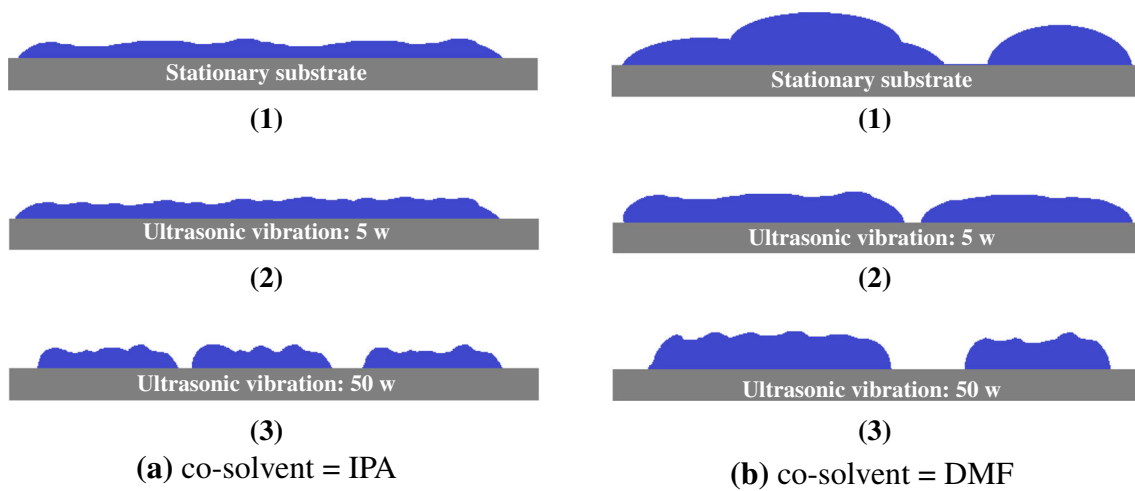


Fig. 7: Schematic summary of the effects of the application of vibration on film wetting/dewetting observed from spray-on PEDOT:PSS films formed in a precursor solution with IPA as the co-solvent (a), and in DMF as the co-solvent (b). The surface features simply show the random film roughness

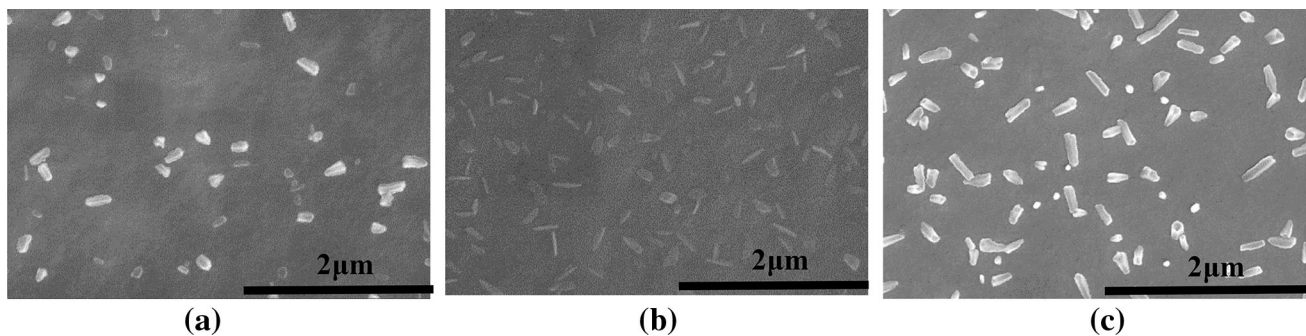


Fig. 8: SEM images of surface crystal size of spray-on PEDOT:PSS films made from a precursor solution containing DMF as the co-solvent at different scenarios: (a) stationary substrate, (b) LPUV, and (c) HPUV

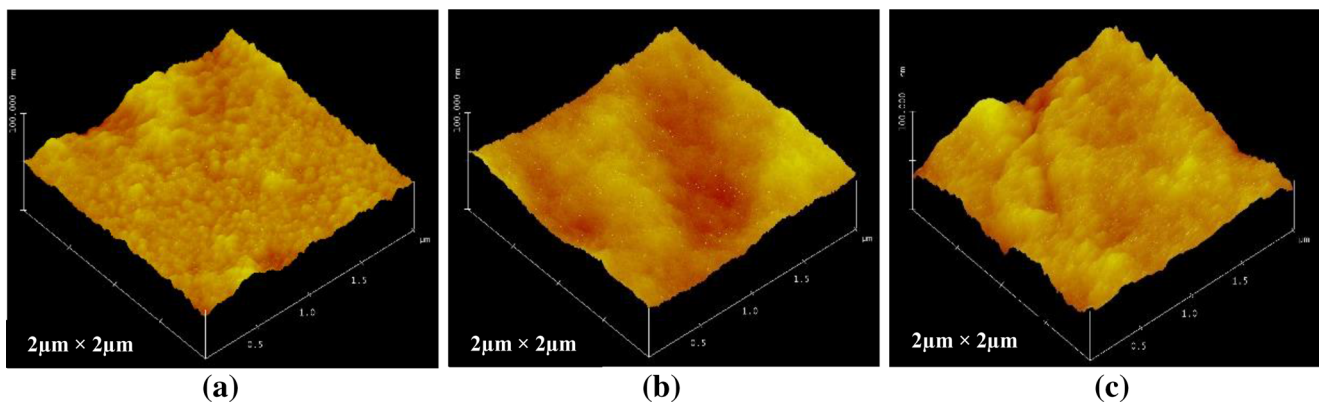


Fig. 9: 3D AFM height images of spray-on PEDOT:PSS thin films made using a precursor solution containing IPA co-solvent, and using 4 spray passes. (a) No substrate vibration, (b) the substrate under the LPUV, and (c) the substrate under the HPUV

phology of perovskite thin film when the LPUV is applied on the substrate. The ultrasonic vibration prohibits the random orientation of crystal growth,

favoring the uniformity in size and morphology of perovskite crystalline matrix. In these experiments, PEDOT:PSS was spun-on FTO. We also incorporated

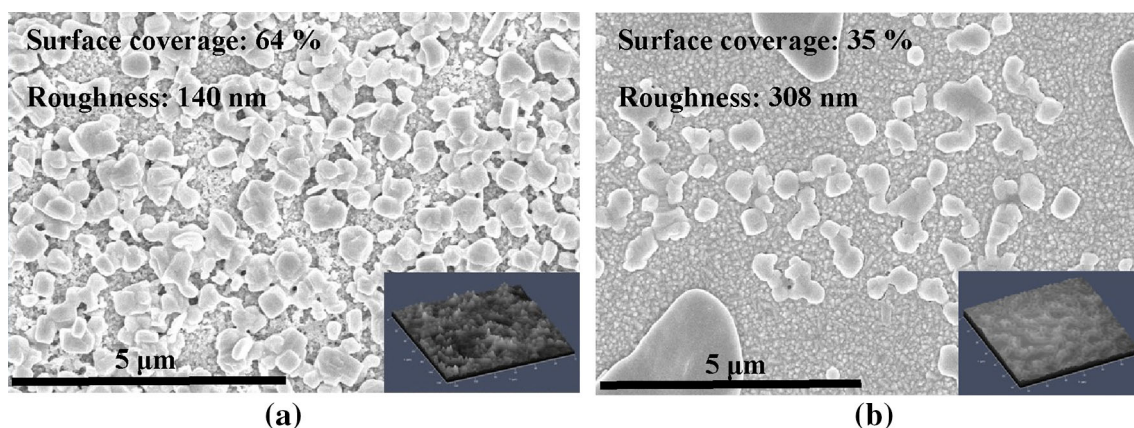


Fig. 10: Optical (inset) and SEM images of perovskite thin films sprayed on FTO/PEDOT:PSS substrates; (a) Vibrating substrate, (b) stationary substrate

a new step in the spinning process, where the as-spun PEDOT:PSS thin film was cured on an ultrasonically vibrating surface (working at 5 W for 10 s), before thermal annealing. We observed a significant improvement in the size, coverage, uniformity, and conductivity of PEDOT:PSS thin films prepared by spin coating followed by substrate vibration.

As mentioned before, the viscosity of the liquid precursor solution is close to that of water, and therefore the stability of the liquid film form upon impingement is governed by the continuity and conventional Navier–Stokes equations for Newtonian flows. As a result of the substrate vibration, the wet film is perturbed. Other forces such as the intermolecular (van der Waals) forces are present as well. The authors are developing the instability analysis behind the wetting/dewetting phenomenon as a result of substrate vibration. Preliminary results on the film instability analysis confirm the effect of both lateral and vertical vibration frequency and power on the film stability in-line with the findings of this work.

Conclusions

The effect of varying the substrate vibration power or amplitude on droplet spreading, wetting, and coating characteristics of thin films made by spray coating was studied. In other similar works, focus has been on the behavior of a single droplet or a train of droplets placed or impinged on a moving or vibrating substrate. In this article, we applied a high frequency ultrasonic vibration (40 kHz) to a substrate during the spray coating process in order to observe the effectiveness of vibration power on the formation of a uniform film using the substrate vibration-assisted spray coating (SVASC), developed and tested in our lab previously. Large-view optical and laser images and close-up SEM and AFM images were used to study the roughness, coverage, and wettability of PEDOT:PSS films fabri-

cated from an aqueous precursor solution treated by IPA and DMF co-solvents. The co-solvents were used to alter the solution wetting properties. The effect of vibration was also studied on perovskite layers deposited atop PEDOT:PSS films.

In conclusion, we found that wetting and uniformity is promoted by imposing moderate vibration on the substrate, using a low-power ultrasonic vibration, at 5 W and 40 kHz. This corresponds to a vertical displacement of 8.7 μm . As a result of the ultrasonic vibration, droplet spreading, coalescence, and surface wetting is promoted. However, the intensity or amplitude of the ultrasonic power must be tuned and kept below a threshold, because it adversely affects the film coverage and roughness. In other words, a high-power ultrasonic vibration (HPUV) may dewet the film (50 W in this study which corresponds to substrate vertical displacement of 57.7 μm). By applying a low-power ultrasonic vibration (LPUV) on a substrate subjected to spray coating, the holes formed as a result of a dewetting mechanism become smaller, even closed and the dewetted areas are repaired. Under favorable conditions, i.e., when IPA is added to the precursor solution as a co-solvent, even and tear-free films are formed compared to the case without using ultrasonic vibration. The application of the HPUV perturbs the film resulting in film rupture and a decrease in the covered area. Under the HPUV, the formed films were found to have higher roughness than the films made under the LPUV, and in some cases the roughness was even higher than the films made without imposing substrate vibration.

We also showed that increasing the number of spray passes or deposition layers results in a higher local roughness in spray-on PEDOT:PSS films, although it helps to have a better coverage, and in some cases a more uniform film in the large scale. Additional passes may lead to stacking of the forthcoming droplets on already impinged droplets, and the creation of a nonuniform film. Therefore, a compromise must be made between having a continuous film without

rupture and dewetting (full coverage) and having a low roughness. Obviously, to have these benefits at the same time, using multiple-pass spraying strategy with applying the LPUV to the substrate is advantageous. The LPUV compensates for the negative effect of the local roughness increase observed when multiple-pass spraying is used. In other words, a uniform and fully covered film may be achieved by using an LPUV strategy combined with several spray passes. The positive effect of applying the LPUV imposed on the substrate was also demonstrated on perovskite layers made atop PEDOT:PSS films, an effect which is very promising in the fabrication of perovskite solar cells.

SEM and AFM images of the samples show that the size and shape of the surface crystal grains and protrusions formed on spray-on PEDOT:PSS thin films change when the substrate is subjected to ultrasonic vibration. Application of the LPUV makes the crystals smaller and thinner, but the HPUV does not make a considerable difference on the surface grain size, when compared to the no-vibration case.

Another major conclusion is that the precursor solution of PEDOT:PSS containing IPA is more wetting than the solution containing DMF. The DMF-contained precursor solution compared to the IPA-contained precursor solution shows higher roughness and lower wetting and coating ability.

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